EVERY FPPF COVERING IS REFINED BY A LOCALLY QUASI-FINITE FLAT FINITE PRESENTATION COVERING

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6465-6474 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 6476-6484, the complete author proof (7 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The pointwise neighbourhood version (lemma-qf-fp-flat-neighbourhood-dominates-fppf, same chapter) producing locally quasi-finite flat finite presentation maps through prescribed points; cited input with the covering assembly written in the retained span.

Source reference. Original lemma 0571: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6437–6449. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00VT: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6484–6484. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 021M: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6484–6484. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{U} = \{S_i \to S\}_{i \in I}$ be an fppf covering of S, see Topologies, Definition 021M. Then there exists an fppf covering $\mathcal{V} = \{T_j \to S\}_{j \in J}$ which refines (see Sites, Definition 00VT) $\mathcal{U}$ such that each $T_j \to S$ is locally quasi-finite.*

Proof. For every $s \in S$ there exists an $i \in I$ such that s is in the image of $S_i \to S$. By Lemma 0571 we can find a morphism $g_s : T_s \to S$ such that $s \in g_s(T_s)$ which is flat, locally of finite presentation and locally quasi-finite and such that g_s factors through $S_i \to S$. Hence $\{T_s \to S\}$ is the desired covering of S that refines $\mathcal{U}$. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.